

Unveiling the Invisible: Interactive Spatial Sensing Transforms Air Flow Measurement

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ABSTRACT

This engaging experience transforms participants into aerodynamicists, enabling them to explore air flow fields firsthand. Utilizing groundbreaking augmented reality visualizations alongside an active learning measurement system, individuals wield a sensor with inside-out tracking, allowing them to navigate and directly uncover the dynamics of air flow. The system generates real-time visualizations of measured air currents, making the usually complex task of evaluating flow fields more accessible. Assisted by a companion artificial intelligence, users are directed to subsequent measurement points, effectively integrating them into the algorithm's process. This innovative approach not only simplifies the examination of air flow fields but also enriches the investigative experience by making participants active contributors to the exploration.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools**; • **Applied computing** → **Physical sciences and engineering**.

KEYWORDS

Augmented Reality, Active Learning, Sequential Design, Environmental Sampling, Spatial Sensing, Human-Machine Teaming

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BACKGROUND

The understanding and measurement of flow fields in air is important in many engineering tasks such as the aerodynamic optimization of cars and aircraft or the design of HVAC systems and industrial processes. Due to the lack of Head-Mounted Display (HMD) technologies, measured real flow fields have not been visualized in situ, and scientific visualizations have often relied on two-dimensional representations of highly three-dimensional phenomena (c.f. [Besançon et al. 2021] and [Lee et al. 2023]). One of the

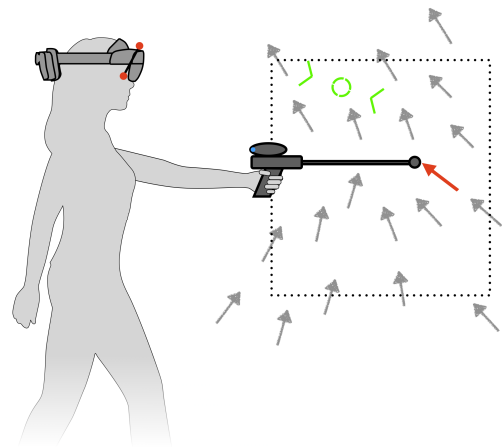
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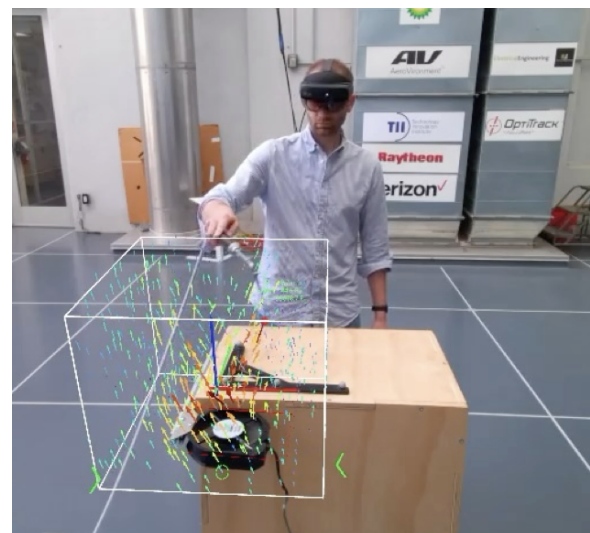
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(a) The user with HMD and handheld with integrated sensor sampling the local field quantity.



(b) Image from the external view of the system. The domain of interest of the algorithm and the next optimal sampling location (green circle).

Figure 1: Machine-guided spatial sensing of flow fields and visualization of current field reconstruction.

earliest works of utilizing XR for in situ flow visualization started in

the early 90s [Bryson 1996; Bryson et al. 1997, 1992]. More involved XR interaction techniques started to emerge a decade later, e.g., [Schkolne 2002] and [Mohr et al. 2019], and a recent advancement in flow field analysis by [Mueller 2017] introduced a method that incorporates a human operator manually guiding a probe to map out a flow field. A clear limitation of their approach is the absence of uncertainty quantification. The system we propose offers direct feedback to the operator, indicating the next optimal point for measurement and builds on the developments of [Humml 2023]. This method facilitates the transfer of expertise from the operator to the algorithm, eliminating the need for the operator to have specific knowledge about the task at hand. The use of augmented reality visualizations significantly enhances the user's hand-eye coordination, playing a crucial role in clearly conveying the location of the next optimal measurement point.

An important aspect of this technique is registering the sensor location and orientation in space. Fixed motion capture systems provide accurate spatial tracking but are cumbersome to set up at new locations. The inside-out tracking of augmented reality headsets is crucial in providing the viewing location and direction of holograms placed in space or the rendering direction in a fully immersive environment. We utilize this tracking to establish a global coordinate of the sensor in space: Infrared cameras on the sensor track an infrared target placed on the operator's head-mounted device, providing a relative position, while the visual-inertial odometry and simultaneous localization and mapping algorithms of the HMD provide a global registration. Merging these two coordinate streams results in a global registration of the sensor. Putting the cameras on the handheld instead of on the headset might seem counterintuitive, but the target on the headset can span a larger area and, thereby, provide more accurate tracking. Also, due to privacy concerns and to avoid inference with their optimized hardware-accelerated tracking algorithms, many HMDs do not allow access to their tracking camera streams for third-party applications.

By leveraging global sensor tracking, it becomes possible to synchronize measured data with their specific locations, thereby creating detailed data instances. These instances are subsequently processed in sequence, either by an algorithm operating on the HMD or externally on a server. This active learning algorithm enhances the user's comprehension of the sampling process. Feedback regarding the current status of the measurement and additional process information is relayed back to the user through their HMD, keeping them informed and engaged throughout the process.

We showcase this technology for air currents; their strong gradients and unsteady characteristics make them difficult to sample and an ideal academic test case, but measuring other spatial fields, such as gas concentrations, temperature, radio frequency signals, etc., is also conceivable in future applications.

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